

Ndumiso Baliti

Exercise 8

$$u(x - \Delta x) = u(x) \left[-\frac{\partial u}{\partial x} \Delta x + O(\Delta x^2) \right]$$

$$\frac{u(x) - u(x - \Delta x)}{\Delta x} = \frac{\frac{\partial u}{\partial x} \cancel{\Delta x} + O(\Delta x^2)}{\Delta x}$$

$$\frac{\partial u}{\partial x} = \frac{u(x) - u(x - \Delta x)}{\Delta x} + O(\Delta x^2)$$

$$u(x_i, t^n) = u_i^n ; \quad u(x_i - \Delta x, t^n) = u_{i-1}^n$$

$$\frac{\partial u}{\partial x} = \frac{u_i^n - u_{i-1}^n}{\Delta x}$$